

INTRODUCTION TO SYNTAX I

Session 2: Syntactic structures have requirements to satisfy

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TODAY'S PLAN

- ◉ Refining hierarchy
- ◉ Smallest units: features
- ◉ Constituents have restrictions and requirements

A CENTRAL RELATION: C-COMMAND

I What are syntactic processes sensitive to?

- ◉ We already know that syntactic processes target constituents, not random strings.
- ◉ They are also sensitive to the **hierarchical relation** between constituents.

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■ A very important one of these hierarchical relation is c-command (c for constituent):

⇒ Node α **c-commands** node β and everything β dominates if α and β are sisters.

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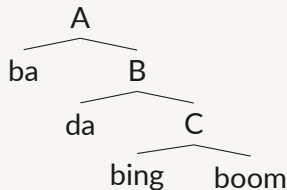
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- Can you name the c-command relations in this tree?
- (C-command will become really important later on!)



I Quick recap

- ⊙ We established that syntax cares about **constituents** (and c-command).
 - ⊙ We have not yet modeled **why syntactic processes are triggered** in the first place.
- ⇒ Syntactic processes do not just happen randomly in this theory of syntax!

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I Recall again the Y-model from yesterday:

- ◉ The mental lexicon contains all **featural and lexical** elements of a language L.
- ◉ The numeration contains certain **featural and lexical** elements selected from the mental lexicon to build the structure at hand.

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- ◉ The mental lexicon contains all **featural and lexical** elements of a language L.
 - ◉ The numeration contains certain **featural and lexical** elements selected from the mental lexicon to build the structure at hand.
- ⇒ Today, we are going to take a closer look at these featural elements.

■ Subject-verb agreement in English

- ⊙ Agreement can be understood as co-variation between two syntactic nodes:

- (1) a. The cat chase**s** the mouse.
b. The cat**s** chase the mouse.

- (2) a. *The cat chase the mouse.
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- ⊙ Based on (2), we could hypothesize that agreement makes reference to the *outer appearance* (morphological form) of words.

⇒ Only one -s is allowed when a third person subject agrees with the verb.

■ **Hypothesis 1: If the subject has an -s, the verb does not and vice versa.**

■ Counterevidence to Hypothesis 1:

- | | | | | | |
|-----|----|-------------------------|-----|----|------------------------------------|
| (3) | a. | The ox eats the hay. | (4) | a. | The child takes piano lessons. |
| | b. | The oxen eat the hay. | | b. | The children take piano lessons. |
| | c. | *The oxen eats the hay. | | c. | *The children takes piano lessons. |

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■ Hypothesis 2: Agreement makes reference to abstract morphosyntactic features.

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Hypothesis 2: Agreement makes reference to abstract morphosyntactic features.

- ⊙ **Morphosyntactic features** are properties of words that syntax is sensitive to.
- ⊙ H2 thus differs from H1 in that it **centers the abstract feature** (which may determine the morphological form), not the form itself.

■ Relevant feature for agreement in English: number

- ⊙ English has plural and singular number.
- ⇒ the number features of the subject and the verb must match.
- ⊙ Subject [pl] → verb [pl]; subject [sg] → verb [sg]
- ⊙ The **morphological form** of these features is context-dependent!

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- ⊙ The **morphological form** of these features is context-dependent!

■ It's the features that matter, not the form.

- ⊙ Plural nouns in English come in various morphological forms: *cat* → -s, *ox* → -en, *child* → -ren, *sheep* → ∅, ...
- ⊙ Is there more evidence for H2? Let's look at the verb *take* cross-linguistically!

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Mandarin

	SG	PL
1	ná	ná
2	ná	ná
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- ⊙ Sensitivity not only to the subject's but the **object's features**, too!

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- ◉ We want a formal system that can derive the simplicity of Mandarin, the complexity of Tundra Nenets and everything in between.
- ◉ Features buy us that system $\Rightarrow$ **Languages vary based on which features of which elements matter** for syntactic processes, such as determining the morphological realization of the verb.

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- ⊙ A feature is a property of a word.
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- ⊙ Example: *cat* [category: noun, number: singular, animacy: animate, ...]

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■ Words are feature bundles containing multiple types of features:

- ① phonological features $\Rightarrow$ /kæt/
 - ② semantic features $\Rightarrow$ {animate}
 - ③ morphosyntactic features $\Rightarrow$ {noun, singular}
- $\Rightarrow$ This information is **stored in the mental lexicon**.

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- ⊙ Are you familiar with other morphosyntactic features?
- ⊙ Gender, case, aspect, tense, mood,...

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■ **These three features are called ϕ -features (*phi*):**

⇒ They are (among) the central features relevant for inflection in many languages.

A NOTE ON GENDER

■ Grammatical gender is mostly arbitrary!

(5) German

- a. das Mädchen: gender: *neuter*, sexus: *female*
- b. der Tisch gender: *masculine*, sexus: -

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- ⊙ Syntax is sensitive to the morphosyntactic feature, i.e. grammatical gender.
- ⊙ **Exception: pronouns!**

- (6) das Mädchen hat sein /ihr Fahrrad geschoben
DET.F girl have.3SG 3SG.POSS.N 3SG.POSS.F bike pushed
'the girl pushed (lit.) its/her bike'

■ Morphosyntactic features can be divided into two groups:

① Inherent features

- ◇ An element X always bears value Y of feature Z, regardless of syntactic context: [Z:Y] is a property of X that is stored in the mental lexicon.
- ⇒ Gender of a noun: *das Mädchen* is always neuter.

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② Assigned features

- ◇ The value of feature Z of element X is determined by the syntactic context (inflection!): the value of Z is not stored in the mental lexicon entry for X.
- ⇒ Gender on an adjective: *klein-es Mädchen, klein-er Tisch*

MORE MORPHOSYNTACTIC FEATURES: CASE

| Case features are assigned based on syntactic context.

- ◉ Informative about the grammatical function of the element that bears them:

(7) *Hungarian*

Fanni-Ø level-et küld-ött Simon-nak.
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- ◉ **Mandarin:** no case
- ◉ **German:** four cases
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- ◉ Large inventories encode locative and directional relations via case:

(8) *Hungarian*

a ház-ban, a ház-ba, a ház-ból
DET house-INESS DET house-ILL DET house-ELA

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Indicates at what time relative to the time of utterance something happened.

- ⊙ **Past, present, future**
- ⊙ Tense can be expressed **synthetically** (within a single verb: *went*)
- ⊙ Or **periphrastically** (via a combination of multiple verbal elements: *will go*)

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I These were only a few morphosyntactic features!

- ⊙ Mood, polarity, voice, definiteness, animacy, ...

TYPES OF CONSTITUENTS

■ Recap: What syntax is sensitive to

- ⊙ **Constituents**, (the hierarchical relation between them) and the **features** they carry.
- ⊙ We have seen how the form of a word adapts to the **features in its context** ⇒ **inflection**.

■ Recap: What syntax is sensitive to

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- ① **Predicates**: have requirements
- ② **Arguments**: satisfy requirements
- ③ **Adjuncts**: only here for the party

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③ Adjuncts:

- ◇ Optional – the sentence is grammatical with or without them.
- ◇ Think of them as embellishments ⇒ *in the cabinet, at 10 in the morning*

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■ **Valency: the number of arguments a predicate needs to express a meaningful event.**

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- ③ **Goal** $\Rightarrow$ the goal/endpoint of the action
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- 3 **Goal** $\Rightarrow$ the goal/endpoint of the action
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- 4 **Experiencer** $\Rightarrow$ the entity experiencing/feeling something
***Trixie** felt hot./**He** was scared.*

CONDITIONS ON θ -ROLE ASSIGNMENT

I θ -role assignment

- ⊙ θ -roles go hand in hand with a predicate's valency.

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■ Theta-criterion:

- ◉ Each θ -role must be assigned to an argument (discharging it does not suffice):

(9) *Arrives.

(10) *Eats the cheese.

- ◉ Each argument is assigned one and only one θ -role.

(11) *Freddie likes. ⇒ Cannot mean *Freddie likes himself*.

■ **There are some apparent counterexamples — you can find them in the appendix.**

I Refining morphosyntactic features

- ⦿ The morphosyntactic features we saw so far determined how the elements in their context inflect $\Rightarrow$ they are called **inflectional features**.

DIFFERENT FEATURES DO DIFFERENT THINGS

I Refining morphosyntactic features

- ⊙ The morphosyntactic features we saw so far determined how the elements in their context inflect $\Rightarrow$ they are called **inflectional features**.
- ⊙ Another type of morphosyntactic feature is the **category feature**: it is an inherent feature stored in the mental lexicon encoding what word category a lexical element belongs to.

I Why do we need category features?

- ⊙ Predicates do not just combine with any argument, they require their arguments to be of a specific category.
- ⊙ This choosiness is called **c-selection** or **subcategorization**.

■ Some predicates require nominal objects:

- (12) a. Barry interrogated the suspect.
b. *Barry interrogated at the suspect.
c. *Barry interrogated screaming.
d. *Barry interrogated whether she knows the suspect.

■ Some others require sentential objects:

- (13) a. Carla admitted that Helen was lovely.
b. *Carla admitted linguistics.

■ Yet others require prepositional objects:

- (14) a. Harry is obsessed with the Beatles.
b. *Harry is obsessed the Beatles.
c. *Harry is obsessed that the Beatles are great.

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■ And a few predicates are happy with an argument with either category feature:

- (15) a. Ringo believes him.
b. Ringo believes in the power of love.
c. Ringo believes that Celine's heart will go on.

C-SELECTION FEATURES

■ C-selection feature notation

- ⊙ We will note c-selection features like this: **[uX]** (**X = variable over category**)
- ⊙ The **number of [uX]-features corresponds to the valency** of the predicate (how many arguments it takes/how many 'places' it has)

■ Each predicate has a category feature and a c-selection feature.

- (16) a. *laugh* [V, uN]
→ intransitive predicate requiring **one nominal argument**
- b. *see* [V, uN, uN]
→ transitive predicate requiring **two nominal arguments**

ARGUMENT STRUCTURE

■ The argument structure specification of predicates could be depicted like this:

- ⊙ It contains the predicate, the θ -roles it assigns and its c-selection features.

Predicate	θ -grid	c-selection	Example
work	agent	[uN]	Claudia works.
love	agent	[uN]	Claudia loves Tina.
	patient	[uN]	
send	agent	[uN]	Claudia sent a letter to Tina.
	patient	[uN]	
	goal	[uP]	

■ The argument structure of a predicate determines:

- ① how many arguments ($\Rightarrow$ valency)
- ② of which category ($\Rightarrow$ c-selection)
- ③ (in which order) **a predicate has to combine with to be *saturated*.**

It also determines:

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It also determines:

- ④ the semantic relation the predicate has with its argument(s) ($\Rightarrow$ θ -role assignment).
- The argument structure of a predicate is **stored in the mental lexicon.**
 - That means it is not computed on the spot and therefore **unpredictable.**

THIS WAS A LOT, BUT IT WILL CARRY US REALLY FAR.

■ It wouldn't surprise me if you thought syntax was a pain in the butt by now.

⊙ **Syntax has a lot of requirements to satisfy!**

- ◇ Lexical elements need to inflect for the abstract features in their context,
- ◇ languages differ wildly regarding in which and to which extent these features matter,
- ◇ predicates need certain arguments and need to assign all of their θ -roles, ...

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- ⊙ This is precisely the point: our language faculty's computational system is **incredibly powerful and creative**, but this power and creativity must also be constrained enough to allow us to **communicate effectively** with one another.

■ Next up: satisfying all of these requirements via structure building!

Ask away!

- (17) a. Dolly donated half a million dollars.
b. Charlie is presenting this week.

■ Making sense of (17)

- ⊙ To perceive these sentences as grammatical, it must be contextually clear what the goal of *donate*/the theme of *present* is.
- ⊙ It could be that the element in question is present in the syntactic structure and simply not realized phonologically.
- ⊙ Not all predicates allow for this, and we will set such cases aside for now.

■ Predicates can also select the semantic properties of their arguments.

- ⊙ For example, some predicates require (in)animate arguments:

- (18) a. #²The wall is running.
b. #The apple is laughing.
c. #Colourless green ideas sleep furiously.

⇒ This is called s-selection (s for semantic).

- ⊙ Syntactically, there is nothing wrong with these examples, but they are semantically ill-formed.

²We mark semantically deviant examples with #.